

PARAINEN UTO, Finland

WMO: 029810

Lat: 59.784N

Lon: 21.368E

Elev: 22

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.9	14.5	5.2	7.2	10.9	9.6	9.0	16.6	42.7	34.1	38.8	34.0	16.6	70	0.814

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	73.8	67.5	71.4	65.9	69.4	64.5	68.8	72.6	66.8	70.6	65.1	68.6	11.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.2	100.2	71.5	65.2	93.3	69.7	63.4	87.4	67.8	32.9	72.5	31.4	70.8	30.0	68.7	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
38.7	34.7	31.4	9.2	75.8	7.5	3.7	3.8	78.4	-0.6	80.6	-4.8	82.6	-10.3	85.3
			8.2	69.3	7.2	2.9	3.0	71.4	-1.2	73.1	-5.2	74.8	-10.5	76.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	30.9	29.0	31.7	38.2	46.4	55.5	63.6	63.7	56.5	47.2	40.2	35.1	(6)
	DBStd	13.14	6.62	7.69	5.45	3.51	4.74	4.24	4.53	3.99	4.24	5.18	5.35	6.14	(7)
	HDD50	3112	592	588	568	355	131	5	0	0	3	114	295	461	(8)
	HDD65	7395	1057	1008	1033	805	575	285	83	72	255	551	745	926	(9)
	CDD50	1263	0	0	0	0	21	170	421	424	199	28	0	0	(10)
	CDD65	72	0	0	0	0	0	0	39	32	1	0	0	0	(11)
	CDH74	67	0	0	0	0	0	0	54	13	0	0	0	0	(12)
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
	WSAvg	16.2	19.3	17.5	16.3	14.1	12.9	13.1	12.9	14.2	16.2	18.2	19.7	20.7	(14)
	PrecAvg	22.4	1.9	1.4	1.3	1.1	1.1	1.7	2.0	2.4	2.4	2.5	2.5	2.2	(15)
(16) Precipitation	PrecMax	27.5	3.1	3.0	2.8	2.1	2.5	3.2	5.0	5.5	7.0	5.2	5.8	3.9	(16)
	PrecMin	17.4	0.3	0.1	0.2	0.2	0.2	0.4	0.1	0.4	0.5	0.2	0.3	0.6	(17)
	PrecStd	2.6	0.7	0.7	0.6	0.5	0.7	0.7	1.4	1.2	1.5	1.1	1.2	0.9	(18)
	PrecStd	2.6	0.7	0.7	0.6	0.5	0.7	0.7	1.4	1.2	1.5	1.1	1.2	0.9	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	40.4	42.0	50.1	64.1	69.4	78.7	75.6	67.4	58.5	50.4	45.9	(19)
		MCWB	41.6	38.5	39.1	44.5	56.4	60.9	70.5	69.5	62.3	56.4	48.8	45.0	(20)
	2%	DB	41.2	39.0	40.3	47.1	59.3	65.6	75.2	72.9	65.3	56.7	49.4	44.5	(21)
		MCWB	40.1	37.7	37.9	42.9	53.5	59.6	68.5	66.7	61.2	54.6	48.2	43.5	(22)
	5%	DB	40.0	38.1	38.8	45.1	55.4	63.5	72.7	71.0	63.7	55.0	48.3	43.3	(23)
		MCWB	38.8	36.9	36.9	41.7	50.8	58.3	67.1	65.4	60.0	53.0	47.0	42.0	(24)
	10%	DB	38.7	37.1	37.6	43.2	52.8	61.7	70.4	69.4	62.1	53.7	46.8	42.1	(25)
		MCWB	37.4	36.0	36.0	40.3	49.1	57.3	65.6	64.4	58.8	51.7	45.3	40.7	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	39.5	39.8	45.3	57.4	62.8	71.9	71.0	64.1	57.4	49.8	45.5	(27)
		MCDB	42.4	39.7	41.1	48.3	62.9	67.0	76.3	74.0	66.4	57.8	50.0	45.6	(28)
	2%	WB	40.4	38.2	38.4	43.6	54.0	60.7	69.6	68.4	62.0	54.9	48.5	43.8	(29)
		MCDB	40.9	38.6	39.6	46.4	58.4	64.2	73.7	71.9	64.3	56.1	49.2	44.1	(30)
	5%	WB	39.1	37.1	37.3	42.1	51.6	59.3	67.8	66.5	60.7	53.6	47.2	42.2	(31)
		MCDB	39.9	37.8	38.4	44.6	54.8	62.7	72.0	69.8	62.8	54.8	48.2	42.9	(32)
	10%	WB	37.6	36.2	36.3	40.7	49.5	57.9	66.1	65.1	59.4	52.1	45.6	40.9	(33)
		MCDB	38.6	37.0	37.3	42.8	52.3	61.1	70.1	68.3	61.5	53.4	46.7	41.9	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.0	5.5	6.4	7.3	6.9	6.7	5.9	5.1	4.3	4.1	4.5	(35)
		MCDBR	4.5	4.1	5.6	8.9	11.1	9.6	8.6	7.1	6.2	4.6	4.1	4.3	(36)
		MCWBR	4.6	4.0	4.6	5.8	6.7	5.6	4.8	4.3	4.6	4.2	4.6	4.6	(37)
	5% WB	MCDBR	4.5	4.0	5.2	8.6	10.6	8.9	7.9	6.4	5.8	4.5	4.1	4.0	(38)
		MCWBR	4.8	4.1	4.4	5.9	6.8	5.6	4.8	4.5	4.7	4.5	4.7	4.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.284	0.317	0.334	0.373	0.368	0.362	0.390	0.382	0.352	0.337	0.312	N/A	(40)
	taud		2.192	2.261	2.318	2.327	2.426	2.485	2.430	2.439	2.504	2.404	2.219	N/A	(41)
	Ebn at Noon		156	216	252	262	273	277	264	258	247	209	141	N/A	(42)
	Edh at Noon		14	22	28	33	32	31	32	30	24	19	13	N/A	(43)
(44) All-Sky Solar Radiation	RadAvg		78	257	755	1310	1803	1977	1847	1443	893	377	105	50	(44)
	RadStd		10	44	84	134	160	123	152	121	102	55	23	7	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	-209	N/A	N/A	N/A

Nomenclature: See separate page